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QUALITY MANAGEMENT SYSTEM PROCEDURE

Title: TRACEABILITY IDENTIFICATION FORMAT SPECIFICATION	
DOCUMENT NUMBER: 960-00098	REVISION: 14
PROCESS OWNER: TECHNICAL OPERATIONS	Page 1 of 11
APPROVERS: HARDWARE ENGINEERING, SUPPLY CHAIN, APQE	

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1. **SCOPE:** This document applies to all components used to manufacture Amazon Robotics (AR) products. Including components, sub-assemblies and final assemblies produced at primary suppliers, sub-suppliers, or Amazon Robotics. Station kits are exempt from these requirements.
2. **PURPOSE:** This document specifies how components and component assemblies are to be identified for traceability purposes. Additionally, the document details what that the traceable identifier is to contain.

3. REFERENCE DOCUMENTS:

960-0015-05	Material Approval Process
960-0075	Engineering Drawing Standards
960-00099	Product Label Specification

4. DEFINITIONS:

- a) AR – Amazon Robotics
- b) Traceability: For a particular manufactured assembly, the ability to verify date of manufacture, vendor or various other attributes of the build cycle
- c) Component Identifier: A numeric or alphanumeric sequence that is assigned to a given component.
- d) Lot: A traceable quantity of parts
- e) Nameplate: A unique traceability type for Amazon Robotics Manufactured finished good assemblies.
- f) AML: Approved Manufacturer List. Maintained in PLM system.

5. RESPONSIBILITIES:

- a) All vendors/suppliers shall comply with this identification format.
- b) Hardware Engineering to collaborate with Technical Operations to define the appropriate traceability type and specify on the product drawing.

6. TRACEABILITY STRATEGY:

- a) The traceability identification levels are broken into groups based on what level of traceability is required. This level is to be specified in the component documentation; which may include but not limited to a Part or Assembly Drawing, Specification Sheet, Purchase Order, AML, or Temporary Deviation Authorization.
- b) Traceability type will be driven by a note on the drawing per the Engineering Drawing Standards document (960-0075).

Typical note: "TRACEABILITY LEVEL: TYPE 08 PER AMAZON ROBOTICS 960-00098"

- c) Traceability types should be chosen to balance risk with respect to cost.
- d) Standardized types should be used wherever possible.
- e) Document 960-00099 defines acceptable methods for applying the required information to the component. These requirements will be validated during the Material Approval Process (960-0015-05).
- f) If a standard type is defined on the assembly print, the characters must be used in the defined order and uninterrupted with no extraneous characters. Additional information (I.E. internal use only), can be appended to the end of the standard code after two dashes (--).

Example encoding work order number with Type 08: AAAYYWWNNNN--WO123456

- g) Unless otherwise stated on the drawing, additional traceability requirements specified on the drawing should be appended to the end specified type, separated by a dash (-).
- h) Parts with the declaration "Material Monitoring Required" in the drawing notes require the supplier to maintain traceability records between the delivered serial/lot number and the test data proving material compliance.

**NOTE: TYPES 00-06 ARE DEPRECATED AND NOT FOR USE ON NEW DESIGNS.
SEE APPENDIX A FOR LEGACY DEFINITIONS.**

7. STANDARDIZED TYPES FOR NEW DESIGNS:

- a) **Type 07 - Nameplate:** individual code that is unique to each AR Drive Unit and shared with license plate requirements for an AR Drive Unit.

Type 07 may be encoded in one of two ways:

Type 07 Option 1: Only to be used on legacy product lines: 1D barcode alongside human readable information. Human readable information shall include Top Level Assembly (TLA) part number, TLA revision number, and drive unit serial number. The 1D barcode encodes the drive unit serial number. The drive unit serial number shall be formatted per the table below.

Type 07 Option 2: Required for all new drive unit product lines. 2D barcode alongside human readable information. Human readable information shall include Top Level Assembly (TLA) part number, TLA revision number, and drive unit serial number. The 2D barcode shall encode the TLA part number, the TLA revision number, and drive unit serial number. Format and delimiters are listed in the table below and must be followed exactly.

- b) **Type 08 – Serialized:** This 12 digit alphanumeric character string consisting of a 3 digit Supplier Code, 2 digit Year of Manufacture, 2 digit Week of Manufacture, and 5 digit Numeric Identifier is a unique identifier that, in combination with the part number, shall not be duplicated under any circumstances. During AR assembly, the part number and serial number are scanned together, and compared to a database to ensure this part number and serial number combination has not been previously used. If a duplicate combination is found, the AR database will prevent use of the part and will generate a negative quality event. Duplicate serial numbers are only acceptable when used in combination with a different part number.

Type 08 may be encoded in one of two ways;

Type 08-Option 1: Only to be used on legacy parts or on special request approved through the MAP. 1D barcode, listing just the SN on a single label, with separate 1D barcodes for Part Number and Revision. Usually with any deviations listed as separate human readable labels added as needed.

Type 08-Option 2: Required for all new products. 2D barcode -Type- Data Matrix, listing SN, PN, Revision (always required) and additional data field for deviations and/or supplier manufacturing data (as agreed upon between Supplier and AR). **Deviation and additional labels can be added as separate labels if it's not possible to incorporate within 2D barcodes.** The additional data fields are available if the Supplier and/or AR see value in increasing transparency to supplier manufacturing data and/or allowing the option to link supplier manufacturing measurements to AR assembly measurements / Field data is beneficial. It is acceptable for the additional data field to be left blank if there is no agreement between the Supplier and AR on additional data sharing. Format and delimiters are listed in the table below and must be followed exactly to allow AR manufacturing tools to properly capture the data during production.

- c) **Type 09 – Lot:** One traceable number to define a quantity of parts produced during a manufacturing run. There is no specific format for the lot number.
- d) **Type 10 – Date Code (Week):** One traceable number to define a quantity of parts produced within a specific week. Date portion can be satisfied with a date wheel with MAP approval.
- e) **Type 11 – Date Code (Month):** One traceable number to define a quantity of parts produced within a specific month. Date portion can be satisfied with a date wheel with MAP approval.
- f) **Type 12 – Julian Date (Casting):** One traceable number to define a quantity of parts produced from a casting setup starting on the specified date.
- g) **Type 13 – Enhanced Traceability:** One 2D barcode consisting of part number, revision, supplier code, date of manufacture with serial number, and a supplier configurable character string allowing product to be traced to critical raw components and manufacturing data. These Supplier configurable string minimum traceability requirements shall be identified on the components mechanical drawing, in the requirement document, or specification and linked up to the Buy Level component. Where a Type 13 Enhanced Traceability 2D barcode exists on a Below the Buy part, Buy level Supplier must capture this data for timely response to future Amazon Robotics query.

Based on the level of traceability specified, the number must be formatted per the following chart:

Type	Level (Type)	Characters
07 Option 1	Name Plate	MM(MM)-BB-YYWW-LLLLL(LL)
Example: G3-NR-1610-1234567 - MM(MM): Model type, can be 2-4 characters, see section 9a - BB: Build Location, can be 2 characters, see section 9b. - YY: 2-digit year of manufacture - WW: 2-digit week of manufacture - LLLLL(LL): License Plate Identifier, can be 5-7 characters, see section 9c.		

07 Option 2	Name Plate	<700-XXXXX-XX(X)\$RR\$MM(MM)-BB-YYWW-LLLLL(LL)\$DDD-DDDDDD\$\$\$>
Example: <700-03500-001\$05\$T1-WB-2333-002100\$DEV-111111&DEV-222222\$\$\$> • 700-XXXXX-XX(X): Top level assembly part number • RR: Revision number of top level assembly part number • MM(MM): Model type, can be 2-4 characters, see section 9a • BB: Build Location, can be 2 characters, see section 9b. • YY: 2-digit year of manufacture • WW: 2-digit week of manufacture • LLLLLL(LL): License Plate Identifier, can be 5-7 characters, see section 9c. • DDD-DDDDDD ○ If one deviation exists, insert a 10-character deviation in a format "DEV-123456" ○ If multiple deviations exist, separate with a field delimiter "&" like in the example above. ○ If no deviations exist, leave two "\$" delimiters for the deviation field • Two reserved/spacer fields (blank at this time, "\$\$") Delimiter: "\$" is the required delimiter between fields and will be used for data analysis. This is required and common to Type 8 which is built into AR software. The beginning and end of the string start with the symbol "<" and end with the ">" to also assist with data analysis and assist software in categorizing 1D vs. 2D barcode type		
08 Option1	Serialized	AAAYYWWNNNNN
Example: 23C162000562 Vendor V002712 – Built 2016 Week 20 – 562nd unit built in week 20		
08 Option2	Serialized	<PPP-PPPPP\$RR\$AAAYYWWNNNNN\$DDD-DDDDDD\$\$\$Add_Data1\$Add_Data2>
Example: <600-01354\$06\$N17190100007\$DEV-111111&DEV-222222\$\$\$Lot#1234\$PO#1234\$Dim1:1.157> • Part Number (PPP-PPPPP): 600-01354 • Revision (RR): 06 • Serial Number (AAAYYWWNNNNN): N17190100007 (Supplier N17, 2019, 7th unit built in the first week) • List of Deviation number(s) against component: • If one deviation exists, insert a 10-character deviation in a format "DEV-123456" • If multiple deviations exist, separate with a field delimiter "&" like in the example above. • If no deviations exist, leave two "\$" delimiters for the deviation field (pre and post deviation delimiters) in addition to the two buffer delimiters (Item 5). • Two reserved/spacer fields (blank at this time, "\$\$") • Additional Supplier Data: Lot#1234\$PO#1234\$Dim1:1.157. This example has three optional fields (Lot#1234, PO#1234, and Dim1.157).		

- Optional data fields can be useful data for the supplier that is chosen to be shared with AR or may be requested by AR.
- Once optional fields are agreed upon between AR and supplier the order of the optional data field shall not change unless AR is notified and approves. This is important since data may be used to aid field issues without the assistance of the supplier and linking the data to the agreed upon data type is critical. In this example the Dim1 reading may be may be trended by AR for improved SPC control and consistency over time is critical to allow for this.
- Additional data field are separated by a delimiter (\$).

Delimiter: "\$" is the required delimiter between fields and will be used for data analysis. "&" is the required delimiter if multiple data entries are available within a single field. This is required and common to Type 8 which is built into AR software.

The beginning and end of the string start with the symbol "<" and end with the ">" to also assist with data analysis and assist software in categorizing 1D vs. 2D barcode type.

As mentioned in **Type 8-Option 2** : Deviations can be added as separate labels if it's not possible to incorporate within 2D barcodes. In that case, part can have more than one label- 1 with P/N, S/N and Rev (<600-01354\$06\$N17190100007\$>) and other with Deviation information.

Examples with acceptable format-

- <600-01354\$06\$N17190100007\$DEV-11111\$\$\$LOT#1234>
- <600-01354\$06\$N17190100007\$DEV-11111\$\$\$LOT#1234\$PO#111>
- <600-01354\$06\$N17190100007\$DEV-11111\$\$\$>
- <600-01354\$06\$N17190100007\$DEV-11111&DEV-22222\$\$\$>
- <600-01354\$06\$N17190100007\$>

09	Lot	AAA-K
Example: 28R-36G5623WO1234 Vendor V002907 - Internal Vendor Lot 36G5623WO1234 (no specific format)		
10	Date Code (Week)	AAA-YYWW
Example: 288-1551 Vendor V002888 – Built in 2015 Week 51		
12	Casting Lot Julian Date	YYDDD
Example: 16006 Lot started on 6 th day of 2016. (Date wheel typical)		
13	Enhanced Traceability	<PPP-PPPPP\$RR\$AAAYYWW\$DDD-DDDDDD\$XXXXXXXXXXXXXXXXXXXXXX>

Example: <420-03759\$06\$N171901\$DEV-111111&DEV-222222\$\$\$ABCDEFGHIJKLMNQRST>

1. Part Number (PPP-PPPP): 420-03759
2. Revision (RR): 06
3. Serial Number : AAAYYWW
Supplier (AAA): N17
Component Manufactured (YYWW): 1901 (First week of 2019)
4. List of Deviation number(s) against component:
 - If one deviation exists, insert a 10 character deviation in a format "DEV-123456".
 - If multiple deviations exist, separate with a field delimiter "&" like in the example above.
 - If no deviations exist, leave two "\$" delimiters for the deviation field (pre and post deviation delimiters) in addition to the two buffer delimiters (Item 6).
Example:<...N171901\$\$\$\$ABCDE...>
5. Two reserve/spacer fields (\$\$, blank at this time)
6. Supplier link to raw Materials/Mfg data using internal code (variable length):
ABCDEFGHIJKLMNQRST

Delimiter: "\$" is the required delimiter between fields and will be used for data analysis. "&" is the required delimiter if multiple data entries are available within a single field. This is required and common to Type 13 which is built into AR software.

The beginning and end of the string start with the symbol "<" and end with the ">" to also assist with data analysis and assist software in categorizing 1D vs. 2D barcode type.

8. COMMON CONTENT:

- a) Supplier Code: Each supplier and that supplier's location will have a unique identification. This code will be generated through the vendor number within the Amazon Robotics ERP system by doing a Base36 conversion of the 6 digit numeric portion of the Vendor code to generate a 3 digit alphanumeric code representing the vendor AND location (VSER Code) and will be communicated to the supplier. Denoted as three (3) digit alphanumeric characters. (e.g. AAA). If conversion generates less than 3 digits pad the leading values with zeros "0" to fill out 3 locations in code.
- b) Year of Manufacture: The year the item was manufactured. Denoted as two (2) digit numeric characters. (e.g. YY)

Year:	2012	2013	2014	2015	2016	2017	2018	and so on...
Code:	12	13	14	15	16	17	18	...

- c) Year Week of Manufacture: The calendar week the item was manufactured. Denoted as two (2) digit numeric characters. (e.g. WK)

Week:	1	2	3	4	5	6	...	10	11	12	...	52
Code:	01	02	03	04	05	06	...	10	11	12	...	52

- d) Numeric Identifier: Is used to identify a specifically traceable unit. This numeric Identifier should reset to "00001" at the beginning of the work week and increment unit by unit to keep track of

individual units made per week (Monday is the first day of the week). Denoted as five (5) digit alphanumeric characters. (e.g. NNNNN)

9. UNIQUE CONTENT (Type 07):

Nameplate type has the following unique content requirements

- a) Model Type: This is defined by Amazon Robotics Process Engineering. Existing product model types are defined by the below table.

TLA Part Number	TLA Description	Model Type
700-00511	DRIVE UNIT G	G1
700-00785	DRIVE UNIT G 2.0	G2
700-00855	DRIVE UNIT G 3.0	G3
700-01045-NA	DRIVE UNIT G 4.0 - NA VERSION	G4
700-01045-EU	DRIVE UNIT G 4.0 - EU VERSION	G4
700-0084	DRIVE UNIT S	S1
700-00895	DRIVE UNIT, S 2.0	S2
700-01055-NA	DRIVE UNIT S 4.0 - NA VERSION	S4
700-01055-EU	DRIVE UNIT S 4.0 - EU VERSION	S4
700-01700	DRIVE UNIT H, NA/JP, ORANGE	H1
700-01700-C2	DRIVE UNIT H, NA/JP, BLUE	H1
700-01800	DRIVE UNIT H, EU, ORANGE	H2
700-01800-C2	DRIVE UNIT H, EU, BLUE	H2
700-01500	DRIVE UNIT P, NA/JP, ORANGE	P1
700-01500-C2	DRIVE UNIT P, NA/JP, BLUE	P1
700-01510	DRIVE UNIT P, EU, ORANGE	P2
700-01510-C2	DRIVE UNIT P, EU, BLUE	P2
700-01502	DRIVE UNIT P-T, NA/JP, ORANGE	PT1
700-01502-C2	DRIVE UNIT P-T, NA/JP, BLUE	PT1
700-01512	DRIVE UNIT P-T, EU, ORANGE	PT2
700-01512-C2	DRIVE UNIT P-T, EU, BLUE	PT2

700-01504-001	DRIVE UNIT, P-L-DU, BLUE, 925MHZ	PL1
700-01504-002	DRIVE UNIT, P-L-DU, BLUE, 433MHZ	PL2
700-01900-NA	DRIVE UNIT, X-SORT, 16IN, NA/JP	X16N
700-01900-EU	DRIVE UNIT, X-SORT, 16IN, EU	X16E
700-04000-001	TLA, DRIVE UNIT, PROTEUS, NA	PRO
700-03500-001	DRIVE UNIT, T-DU, 925MHZ	T1
700-03500-002	DRIVE UNIT, T-DU, 433MHZ	T2
700-03500-101	DRIVE UNIT, TR-DU, 925MHZ	TR1

b) Build Location(s): Defined by this Table

Build Location	Identifier Code
Amazon Robotics 300 Riverpark Drive North Reading, MA 01864	NR
Amazon Robotics 50 Otis Street Westborough, MA 01581	WB

c) License Plate Identifier: Is used to identify a specifically traceable unit. License Plate Identifier will be one of the following options, based on product definition:

- i. Start at 0000001 and sequence serially until 9999999, Example: 1234567
- ii. Start at aa001 and sequence serially until zz999. Example: ab123

APPENDIX A – LEGACY TRACEABILITY TYPES

Type 00 – **No Traceability Required**

Type 01 - **Lot Code**: A period of manufacture that can be of varying length, quantity and is a steady state production run.

Type 02 - **Date of Manufacture**: Tracking a period of manufacture not more than twenty four (24) hours.

Type 03 - **Date Code**: Specific to shift or machine on a given date.

Type 04 - **Batch code**: Unique production series with a defined quantity, period or process.

Type 05 - **Serialized**: Individual code that is unique to each part.

Type 06 - **Date Limited**: Tracking a period of manufacture to within one week.

Level (Type)	Characters	Example
Serialized	AAAYYWWNNNNNN	OC3122010369
Batch Code	AAAYYWWNNNNNN	MC2114500089
Date Code	AAAYYWWDNNNNNN	VUL1135301B04
Date of Mfg	AAAYYWWD	OC112492
Lot Code	AAAYYWWNNNNNN	KV1120500129
Date Limited	YYWW	1310

REVISION HISTORY

Rev.	ECO #	Description of Change	Sections Affected	Originator/Department
01	ECO-001953	Initial Revision	All	M. Sandman/Manufacturing Technical Ops
02	ECO-003044	Update to supplier code	7.a	M. Sandman/Manufacturing Technical Ops
03	ECO-003337	Update Proprietary, change supplier code to 3 digit Syteline VSER code. Remove product code from serialized labels. Separate Rev and SN labels. Add new "Date Limited" Type (06).	1, 7,8	P. Groudass/Manufacturing Tech Ops.
04	ECO-003783	Added "Nameplate", chg'd "six (6)" to "seven (7)", added, added "(Type 01 to Type 06)", added "Content (Type 07)", added example for 700-0063	4.h., 6.a., 6.a.vii & 6.d., 7., 8., Appendix A	P Hess/Manufacturing Tech Ops
05	ECO-005353	Updated Numeric Identifier definition for clarification	Section 7.a.v	S. Savage/Mechanical Tech Ops
06	ECO-006852	Updated scope to exclude station kits.	1.	M. Anderson/ Tech Ops
07	ECO-009123	Complete rewrite. Deprecated original types 1-6 and moved to Appendix A.	All	M. Anderson/ Tech Ops
08	ECO-011376	Updated to include language regarding duplicate serial numbers.	7.b	B. Quaglieri / Supplier Quality
09	ECO-011730	Updated language to state character length requirement. Corrected language around duplicate serial numbers.	7b	B. Quaglieri / Supplier Quality
10	ECO-014447	Added Type 8, Option 2 and Type 13	7	B. Harty / Technical Operations
11	ECO-018788	Added requirement of 2D barcodes, type- Data Matrix and updated to allow separate deviation labels	7b	P. Narayan, P. Muttha/ APQE
12	ECO-020072	Added examples of acceptable label formats for Type 8, option 2	7b	P. Narayan, P. Muttha/ APQE
13	ECO-020396	Added Type 07 Option 2, clarified data fields in Type 07, added Model type table, added Westborough, added LPN options.	7a, 7b, 9a, 9b, 9c	J. Shipley / AME & R. Shultz / Process Engineering
14	ECO-021335	Updated Type 07 Option 1 and section 9c to account for increase of license plate identifier (serial number) digits from 6 to 7	7a, 7b, 9c	S. Oliver / AR HW Mechanical